The Law of Quadratic Reciprocity

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Motivation

When working in the ring of the integers mod a prime, $\mathbb{Z}/p\mathbb{Z}$ it is natural to wonder whether or not a number is a perfect square, or more formally,

> For some $a \in \mathbb{Z}/p\mathbb{Z}$, does there exist an $x \in \mathbb{Z}/p\mathbb{Z}$ such that
> $x^2 \equiv a \pmod{p}$?

For small values of a and p this question can be pretty trivial. For example, if one were to ask if 2 is a square mod 7 we could simply check every element (equivalence class) in $\mathbb{Z}/7\mathbb{Z}$:

$$0^2 = 0 \not\equiv 2 \pmod{7} \quad 1^2 = 1 \not\equiv 2 \pmod{7}$$
$$2^2 = 4 \not\equiv 2 \pmod{7} \quad 3^2 = 9 \equiv 2 \pmod{7}$$

so, 2 **is** a square mod 7! But what if I asked if 11 was a square mod 71 or if 41 was a square mod 67? It would be quite annoying to calculate the squares of every element of $\mathbb{Z}/71\mathbb{Z}$ and check congruence - this is where the **Law of Quadratic Reciprocity** comes in!

Background (Definitions & Theorems)

- A **ring** is a triple $(R, +, \times)$ which has two operations (often referred to as addition and multiplication) such that
  1. $(R, +)$ is an abelian group with identity 0
  2. $\times$ is an associative, binary operation on R with identity 1
  3. Multiplication distributes over addition
- An **algebraic integer** is a complex number ω that is the root of a polynomial $x^n + a_1x^{n-1} + \cdots + a_n = 0$ where $a_1, a_2, \ldots, a_n \in \mathbb{Z}$.
  - **Theorem:** *The set of algebraic integers forms a ring,* Ω
  - Congruence in Ω follows all the standard properties of congruence in $\mathbb{Z}$ with the added property that $(\omega_1 + \omega_2)^p \equiv \omega_1^p + \omega_2^p \pmod{p}$ for $\omega_1, \omega_2 \in \Omega$ and $p \in \mathbb{Z}$ prime (proved by expanded with the binomial theorem).
- The **Legendre Symbol** is defined for $p \in \mathbb{Z}$ a prime and $a \in \mathbb{Z}$ as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square mod } p \\ -1 & \text{if } a \text{ is not a square mod } p \\ 0 & \text{if } p \mid a \end{cases}$$

- The Legendre Symbol has 3 main important properties
  1. $a^{(p-1)/2} \equiv (a/p) \pmod{p}$
  2. $(ab/p) = (a/p)(b/p)$
  3. If $a \equiv b \pmod{p}$ then $(a/p) = (b/p)$
- An n-th **root of unity** is any value is a solution to the equation $x^n - 1 = 0$
  - We will define $\zeta = \mathbf{e^{2\pi i/p}}$ which is a p-th root of unity
- **Lemma (Orthogonality of Characters):** *The sum* $\sum_{t=0}^{p-1} \zeta^{at}$ *equals* p *is* $a \equiv 0 \pmod{p}$ *and is* 0 *otherwise*

Quadratic Gauss Sums

A **quadratic gauss sum**, g_a is defined as

$$g_a = \sum_{t=0}^{p-1} \left(\frac{t}{p}\right) \zeta^{at}$$

with g_1 commonly notated as g. Interestingly, we have the equivalence that

$$g_a = \left(\frac{a}{p}\right) g.$$

We have the following important lemma surrounding gauss sums:

Lemma

$$g^2 = (-1)^{(p-1)/2} p$$

We can prove this lemma by considering the $\sum_{a=0}^{p-1} g_a g_{-a}$ in two different ways, with our result following from their equality. First we use the identity $g_a = (a/p)g$,

$$g_a g_{-a} = \left(\frac{a}{p}\right)\left(\frac{-a}{p}\right) g^2$$

and since the Legendre Symbol is multiplicative,

$$g_g g_{-a} = \left(\frac{-a^2}{p}\right) g^2 = \left(\frac{-1}{p}\right)\left(\frac{a^2}{p}\right) g^2.$$

If $a \not\equiv 0 \pmod{p}$ then a^2 is a quadratic residue so $(a^2/p) = 1$. Thus,

$$\sum_{a=0}^{p-1} g_a g_{-a} = \left(\frac{-1}{p}\right)(p-1)g^2$$

since there are $p-1$ nonzero values of a mod p. Next, we start by rewriting the product $g_a g_{-a}$ using the definition of a Gauss Sum:

$$g_a g_{-a} = \left(\sum_{x=0}^{p-1} \left(\frac{x}{p}\right) \zeta^{ax}\right)\left(\sum_{y=0}^{p-1} \left(\frac{y}{p}\right) \zeta^{-ay}\right) = \sum_x \sum_y \left(\frac{x}{p}\right)\left(\frac{y}{p}\right) \zeta^{a(x-y)}$$

Summing over a and applying the orthogonality of characters then gives

$$\sum_a g_a g_{-a} = \sum_a \sum_x \sum_y \left(\frac{x}{p}\right)\left(\frac{y}{p}\right) \zeta^{a(x-y)} = (p-1)p.$$

So,

$$\left(\frac{-1}{p}\right)(p-1)g^2 = (p-1)p \Rightarrow g^2 = \left(\frac{-1}{p}\right) p$$

which was to be shown. □

The Law of Quadratic Reciprocity

We now formally state the Law of Quadratic Reciprocity:

Theorem (Quadratic Reciprocity)

For two odd primes p and q with $p \neq q$

$$\left(\frac{p}{q}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} \left(\frac{q}{p}\right)$$

Notice the following:

1. if p **or** $q \equiv 1 \pmod{4}$ then
$$(-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} = 1$$
2. if p **and** $q \equiv 3 \pmod{4}$ then
$$(-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} = -1.$$

As well, it can be shown that for any p then

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} \quad \text{and} \quad \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

Proof

Consider $p^* = (-1)^{(p-1)/2} p = g^2$, and another odd prime q such that $q \neq p$. We will be working with congruences mod q in the ring of algebraic integers:

$$g^{q-1} = \left(g^2\right)^{(q-1)/2} = p^{*(q-1)/2} \equiv \left(\frac{p^*}{q}\right) \pmod{q}$$
$$\Rightarrow g^q \equiv \left(\frac{p^*}{q}\right) g \pmod{q}$$

Rewriting g^q using the definition of the Gauss sum gives

$$g^q = \left(\sum \left(\frac{t}{p}\right) \zeta^t\right)^q \equiv \sum \left(\frac{t}{p}\right)^q \zeta^{qt} \equiv g_q \pmod{q}$$

Recalling that $g_a = (a/p)g$, we now have that $g^q \equiv g_q \equiv (q/p)g \pmod{q}$, thus

$$\left(\frac{q}{p}\right) g \equiv \left(\frac{p^*}{q}\right) g \pmod{q}$$

If we multiply both sides by g and apply the fact that $g^2 = p^*$ we have

$$\left(\frac{q}{p}\right) p^* \equiv \left(\frac{p^*}{q}\right) p^* \pmod{q}$$
$$\Rightarrow \left(\frac{q}{p}\right) \equiv \left(\frac{p^*}{q}\right) \pmod{q}$$
$$\Rightarrow \left(\frac{q}{p}\right) = \left(\frac{p^*}{q}\right)$$

Which was to be shown. □

Example

Earlier we wondered if 11 was a square, or quadratic residue, mod 71. Using the law of quadratic reciprocity we can figure this out:

$$\left(\frac{11}{71}\right) = (-1)^{(11-1/2)(71-1/2)} \left(\frac{71}{11}\right)$$
$$= -\left(\frac{5}{11}\right) = -(-1)^{(5-1/2)(11-1/2)} \left(\frac{11}{5}\right)$$
$$= -\left(\frac{1}{5}\right) = -(-1)^{5-1/2} = -1$$

Therefore, 11 **is not** a square mod 71. This also implies that 71 **is** a square mod 11.

References

K. Ireland and M. Rosen *A Classical Introduction to Modern Number Theory*

Future Work

Going forward, I hope to continue to learn more about quadratic reciprocity as well as the extensions such as cubic, quartic, and bi-quadratic reciprocity. I also would like to learn more about how quadratic reciprocity can be used to help factor in the Gaussian Integers $\mathbb{Z}[i]$.